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| | **Method Validation for Calorific Value Determination** | **Document no: AS002MV** |
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| ***Compiled by:*** | **Sifiso Nyambi - Laboratory Manager** |  |
| ***Approved by:*** | **Nonhlanhla Msibi -Operations Manager** |  |

Measurement Method: ISO 1928

Reference Material (RM): GA CRM AS002

Date of Characterization: May 2025

Analyst: Keitumetse

Number of Replicates: 10 units × 2 replicates

## 1. Introduction

This report presents the validation of the method applied for the determination of calorific value in solid fuel samples, in accordance with ISO 1928. The objective of validation is to confirm that the method provides reliable, accurate, and precise results under repeatability, reproducibility, and robust conditions.

**Key parameters assessed include:**

- Precision (repeatability and reproducibility)
- Accuracy (trueness and recovery)
- Robustness/ruggedness (temperature effect)
- Homogeneity of the reference material
- Measurement uncertainty

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## 2. Methodology

- Sample mass range: 0.5000–0.5099 g
- Replicates: 8–10 replicate determinations per analyst per condition
- Assigned value (reference): 24.07 MJ/kg (based on benzoic acid calibration AV = 26.43 MJ/kg)
- Acceptance criteria:
  - Difference between measured and assigned value ≤0.30 MJ/kg
  - Recovery: 98–102%
  - Precision (RSD): ≤1%

Data was collected from three analysts under normal and elevated temperature conditions, followed by statistical analysis (t-tests, ANOVA, homogeneity testing, and uncertainty estimation).

## 3. Results

### 3.1 Precision (Repeatability and Reproducibility)

| Analyst | Average (MJ/kg) | SD | RSD (%) | Absolute Difference from AV | Recovery (%) |
|---|---|---|---|---|---|
| 1 | 23.93 | 0.09 | 0.39 | 0.14 | 99.42 |
| 2 | 23.77 | 0.05 | 0.21 | 0.30 | 98.76 |
| 3 | 23.93 | 0.05 | 0.21 | 0.14 | 99.42 |

RSD values are all <0.5%, confirming excellent repeatability and reproducibility. Absolute differences ≤0.30 MJ/kg, within acceptance. Recoveries between 98.76–99.42%, within acceptance.

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Conclusion: Precision and accuracy requirements are met.

### 3.2 Ruggedness / Robustness (Temperature Effect)

ANOVA was used to compare results under normal lab conditions (19–26 °C) and elevated conditions (27.56 °C).

- Normal average: 23.93 MJ/kg (variance 0.0088)
- Elevated average: 23.88 MJ/kg (variance 0.0052)
- F-stat = 1.72 < F-critical = 4.60; p = 0.21 > 0.05

  Conclusion: No statistically significant difference between conditions. Method is robust to small environmental temperature changes.

### 3.3 Homogeneity Assessment

10 units × 2 replicates were analysed. ANOVA showed:

- Fcalc = 0.68 < Fcrit = 3.02
- No significant between-unit variation detected.
- Homogeneity confirmed.

### 3.4 Measurement Uncertainty

#### 3.4.1 Sources of Uncertainty

| Source | Symbol | Type | Std. Uncertainty (U) | Distribution | Divisor | Sensitivity | U² |
|---|---|---|---|---|---|---|---|
| Repeatability | U1 | A | 0.12 | Normal | 1 | 1 | 0.0144 |
| Balance calibration | U2 | B | 0.0157 | Rectangular | √3 | 1 | 0.00025 |
| Homogeneity | U3 | A | 0.1083 | Normal | 1 | 1 | 0.01174 |

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|---|---|---|---|---|---|---|---|
| Stability | U4 | B | NA | Rectangular | √3 | 1 | - |
| PT Characterisation | U5 | B | NA | Normal | 1 | 1 | - |

### 3.4.2 Combined and Expanded Uncertainty

- Combined Uncertainty (Uc): 0.14 MJ/kg
- Expanded Uncertainty (U, k=2): 0.28 MJ/kg
- Coverage probability: 95%

## 4. Discussion

- Precision: Both intra-analyst (repeatability) and inter-analyst (reproducibility) results show RSD <0.5%, which is well within acceptance limits.
- Accuracy: Absolute differences ≤0.30 MJ/kg and recoveries close to 100% demonstrate trueness.
- Ruggedness: Method performance is not significantly affected by moderate environmental temperature variations.
- Homogeneity: CRM units show no significant between-unit differences, confirming reliability. Measurement uncertainty: The final expanded uncertainty of 0.28 MJ/kg is consistent with typical performance for ISO 1928 calorific value determination.

| **Parameter** | **Justification for Omission** | **Data Source / Reference** |
|---|---|---|
| **Limit of** | Technically not applicable. CV measures the bulk | **ISO 1928 /** |

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| **Parameter** | **Justification for Omission** | **Data Source / Reference** |
|---|---|---|
| **Detection (LOD)** | energy content of a sample. A sample at the "limit of detection" would have zero energy, which is outside the scope of solid fuel testing. | **AS002MV Section 1** |
| **Limit of Quantitation (LOQ)** | Not applicable. The method is used for quantifying energy in MJ/kg, not for trace analysis of chemical constituents. | **ISO 1928 / AS002MV Section 1** |
| **Linearity** | The calorimeter is calibrated using Benzoic Acid to determine the water equivalent (heat capacity) of the system. The heat release vs. temperature rise relationship is fundamentally linear within the operating range. | **ISO 1928 / AS002MV Section 4** |
| **Selectivity / Specificity** | The method measures "Gross" calorific value, representing total energy release upon combustion. Selecting one specific energy-producing component over another is not required by the standard. | **ISO 1928 / AS002MV Section 1** |

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| Effective date | Rev status | Next review date |
|---|---|---|
| 02/2026 | 01 | 02/2029 |

| Parameter | Justification for Omission | Data Source / Reference |
|---|---|---|
| **Analyte Stability** | Analyte stability is a function of sample storage and preservation (preventing oxidation/moisture loss) rather than a variable of the ISO 1928 measurement procedure itself. | **ISO 1928 / AS002MV Section 2** |

## 5. Conclusion

- The method validated in this study is:
- Precise (RSD <0.5%)
- Accurate (recoveries 98.8–99.4%)
- Robust (insensitive to small environmental changes)
- Homogeneous (CRM units consistent)
- Traceable and reliable (expanded uncertainty 0.28 MJ/kg, 95% CL)

**Therefore, the method meets ISO 1928 requirements and is fit for purpose for the determination of calorific value in solid fuel samples.**

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## 6. Final Reporting Statement

The calorific value determination by ISO 1928 method has an expanded uncertainty of 0.28 MJ/kg (k=2, 95% confidence level). All results are traceable to SI units through calibrated reference materials and instruments.

## 7. Revision

| Revision no | Content changed | Reason for revision | Date changed |
|---|---|---|---|
| 01 | No contents changed, additional required information. | Added table for justification of omitted performance characteristics as a requirement | 02/2025 |
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